4.1.1 Show that intrinsic sum of for E_k in

(related to 4.2.1) (4.2) $E_k(\tau) = \frac{1}{2} \sum_{\gamma \in P_+ \backslash SL_2(\mathbb{Z})} j(\gamma,\tau)^{-k}$ agrees with

(4.1) $E_k(\tau) = \frac{1}{2} \sum_{\substack{(c,d)\in\mathbb{Z}^2 \\ \gcd(c,d)=1}} \frac{1}{(c\tau+d)^k}$.

Pf: as $j(\gamma,\tau)^{-k} = \frac{1}{(c\tau+d)^k}$, we need to show

$\gamma \in P_+ \backslash SL_2(\mathbb{Z}) \iff \gcd(c,d)=1$.

Recall $P_+ = \{\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}, n \in \mathbb{Z}\}$. then $P_+ \backslash SL_2(\mathbb{Z})$

$= \{\begin{pmatrix} a+cn & b+dn \\ c & d \end{pmatrix} : n \in \mathbb{Z}\}/\sim$ where $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \sim \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$ iff $\begin{cases} c=c' \\ d=d' \end{cases}$

Thus $\forall \gamma \in P_+ \backslash SL_2(\mathbb{Z})$. ~~$j(\gamma,\tau) =$~~ as a representative, $\gcd(c,d)=1$.

4.1.2 Confirm formula (4.3)

$$\dim(\mathcal{E}_k(\Gamma)) = \begin{cases} ① \ \varepsilon_\infty & k \geq 4 \text{ even} \\ ② \ \varepsilon_\infty^{reg} & k \geq 3 \text{ odd}, -I \notin \Gamma \\ ③ \ \varepsilon_\infty - 1 & k=2 \\ ④ \ \varepsilon_\infty^{reg}/2 & k=1, -I \notin \Gamma \\ ⑤ \ 1 & k=0 \\ ⑥ \ 0 & k<0, \text{ or } k>0 \text{ odd } -I \in \Gamma \end{cases}$$

Pf: $\dim(\mathcal{E}_k(\Gamma)) = \dim(\mathcal{M}_k(\Gamma)) - \dim(S_k(\Gamma))$

$$\dim(\mathcal{M}_k(\Gamma)) = \begin{cases} (k-1)(g-1) + \lfloor \frac{k}{3} \rfloor \varepsilon_3 + \frac{k}{2} \varepsilon_\infty^{reg} + \frac{k-1}{2} \varepsilon_\infty^{irr} & k \geq 3 \text{ odd} \\ (k-1)(g-1) + \lfloor \frac{k}{4} \rfloor \varepsilon_4 + \lfloor \frac{k}{3} \rfloor \varepsilon_3 + \frac{k}{2} \varepsilon_\infty & k \geq 2 \text{ even} \\ \geq \frac{1}{2} \varepsilon_\infty^{reg} & k=1 \\ 1 & k=0 \\ 0 & k<0 \end{cases}$$

$$\dim(S_k(\Gamma)) = \begin{cases} (k-1)(g-1) + \lfloor \frac{k}{4} \rfloor \varepsilon_4 + \lfloor \frac{k}{3} \rfloor \varepsilon_3 + (\frac{k}{2}-1) \varepsilon_\infty & k \geq 4 \text{ even} \\ (k-1)(g-1) + \lfloor \frac{k}{3} \rfloor \varepsilon_3 + \frac{k-2}{2} \varepsilon_\infty^{reg} + \frac{k-1}{2} \varepsilon_\infty^{irr} & k \geq 3 \text{ odd} \\ g & k=2 \\ \dim(\mathcal{M}_1(\Gamma)) - \frac{1}{2} \varepsilon_\infty^{reg} & k=1 \\ & k \leq 0 \end{cases}$$

$\dim(\mathcal{E}_{\geq 4 \text{ even}}(\Gamma)) = \varepsilon_\infty$.

$\dim(\mathcal{E}_{\geq 3 \text{ odd}}(\Gamma)) = \varepsilon_\infty^{reg}$

$\dim(\mathcal{E}_2(\Gamma)) = \varepsilon_\infty - 1$.

$\dim(\mathcal{E}_1(\Gamma)) = \frac{1}{2} \varepsilon_\infty^{reg}$

$\dim(\mathcal{E}_0(\Gamma)) = 1$

$\dim(\mathcal{E}_{<0}(\Gamma)) = 0$.